SKILLSHEET

# Mental percentages

## Finding 10% of a number

Finding 10% or $\frac{1}{10}$ of a number is the same as dividing that number by 10. To find 10% of a number, we move the decimal point one place to the left.

### Example 1

Find:

**a** 10% of 850 **b** 10% of 1756 **c** 10% of 34

### Solution

**a** 10% of 850 = 850 ÷ 10 = 85.0 = 85

**b** 10% of 1756 = 1756 ÷ 10 = 175.6 = 175.6

**c** 10% of 34 = 34 ÷ 10 = 3.4 = 3.4

### Finding 5% of a number

Since 5% is half of 10%, to find 5% we divide the number by 10, then find half of it. To find 5% of a number, we move the decimal point one place to the left, then divide by 2.

### Example 2

Find:

**a** 5% of 360 **b** 5% of $22 **c** 5% of 70

### Solution

**a** 10% of 360 = 36
So 5% of 360 = 36 ÷ 2 = 18

**b** 10% of $22 = $2.20
So 5% of $22 = $2.20 ÷ 2 = $1.10

**c** 10% of 70 = 7
So 5% of 70 = 7 ÷ 2 = 3.5

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## Exercise

**1** Find the following percentages without using a calculator.

**a** $10\% \times 24$ **b** $10\% \times 212$ **c** $10\% \times \$549$ **d** $10\% \times \$45$

**e** $5\% \times 880$ **f** $5\% \times 104$ **g** $5\% \times \$41$ **h** $10\% \times \$260$

**i** $10\% \times 1925$ **j** $5\% \times 270$ **k** $5\% \times \$64$ **l** $5\% \times \$182$

# Finding $2\frac{1}{2}\%$ of a number

Since $2\frac{1}{2}\%$ is half of 5%, to find $2\frac{1}{2}\%$ we find 5% of the number, then find half of it. So, to find $2\frac{1}{2}\%$ of a number, we move the decimal point one place to the left, then divide by 2, then divide by 2 again.

## Example 3

Find:

**a** $2\frac{1}{2}\%$ of 160 **b** $2\frac{1}{2}\%$ of \$86 **c** $2\frac{1}{2}\%$ of 548

## Solution

**a** 10% of 160 = 16

So 5% = 16 ÷ 2 = 8

So $2\frac{1}{2}\% = 8 \div 2 = 4$

$2\frac{1}{2}\% \times 160 = 4$

**b** 10% of \$86 = \$8.60

So 5% = \$8.60 ÷ 2 = \$4.30

So $2\frac{1}{2}\% = \$4.30 \div 2 = \$2.15$

$2\frac{1}{2}\% \times \$86 = \$2.15$

**c** 10% of 548 = 54.8

So 5% = 54.8 ÷ 2 = 27.4

So $2\frac{1}{2}\% = 27.4 \div 2 = 13.7$

$2\frac{1}{2}\% \times 548 = 13.7$

# Finding 20% of a number

Since 20% is twice 10%, to find 20% we find 10% of the number, then double it. So, to find 20% of a number, move the decimal point one place to the left, then double the result.

## Example 4

Find:

**a** 20% of 45 **b** 20% of \$128 **c** 20% of 300

## Solution

**a** 10% of 45 = 4.5
So 20% of $45 = 4.5 \times 2 = 9$

**b** 10% of \$128 = \$12.80
So 20% of \$128 = \$12.80 × 2 = \$25.60

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**c** 10% of 300 = 30
So 20% of 300 = 30 × 2 = 60

## Exercise

**2** Find the following percentages without using a calculator.

**a** $2\frac{1}{2}\% \times 200$ **b** $2\frac{1}{2}\% \times 480$ **c** $2\frac{1}{2}\% \times \$184$ **d** $2\frac{1}{2}\% \times \$660$

**e** $20\% \times 720$ **f** $20\% \times 25$ **g** $20\% \times \$545$ **h** $20\% \times \$390$

**i** $2\frac{1}{2}\% \times 250$ **j** $20\% \times 134$ **k** $2\frac{1}{2}\% \times \$120$ **l** $2\frac{1}{2}\% \times \$56$

# Finding 50% of a number

Finding 50% or $\frac{1}{2}$ of a number is the same as dividing that number by 2.

## Example 5

Find:

**a** 50% of 74 **b** 50% of $245

## Solution

**a** 50% of 74 = 74 ÷ 2 = 37

**b** 50% of $245 = $245 ÷ 2 = $122.50

# Finding 25% of a number

Finding 25% or $\frac{1}{4}$ of a number is the same as dividing that number by 4. To find 25% of a number, we divide by 2 (to get 50%), then divide by 2 again.

## Example 6

Find:

**a** 25% of 850 **b** 25% of $133

## Solution

**a** 50% of 850 = 850 ÷ 2 = 425
25% of 850 = 425 ÷ 2 = 212.5

**b** 50% of $133 = $133 ÷ 2 = $66.50
25% of $133 = $66.50 ÷ 2 = $33.25

## Exercise

**3** Find the following percentages without using a calculator.

**a** $50\% \times 70$ **b** $50\% \times 45$ **c** $25\% \times 144$ **d** $25\% \times 22$

**e** $25\% \times \$62$ **f** $50\% \times \$27$ **g** $25\% \times 108$ **h** $50\% \times 17$

**i** $50\% \times \$24.50$ **j** $25\% \times \$680$ **k** $25\% \times \$96$ **l** $50\% \times \$185$

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## Answers

| | | | | |
|---|---|---|---|---|
| **1** | **a** 2.4 | **b** 21.2 | **c** $54.90 | **d** $4.50 |
| | **e** 44 | **f** 5.2 | **g** $2.05 | **h** $26 |
| | **i** $192.5 | **j** 13.5 | **k** $3.20 | **l** $9.10 |
| **2** | **a** 5 | **b** 12 | **c** $4.60 | **d** $16.50 |
| | **e** 144 | **f** 5 | **g** $109 | **h** $78 |
| | **i** 6.25 | **j** 26.8 | **k** $3 | **l** $1.40 |
| **3** | **a** 35 | **b** 22.5 | **c** 36 | **d** 5.5 |
| | **e** $15.50 | **f** $13.50 | **g** 27 | **h** 8.5 |
| | **i** $12.25 | **j** $170 | **k** $24 | **l** $92.50 |

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